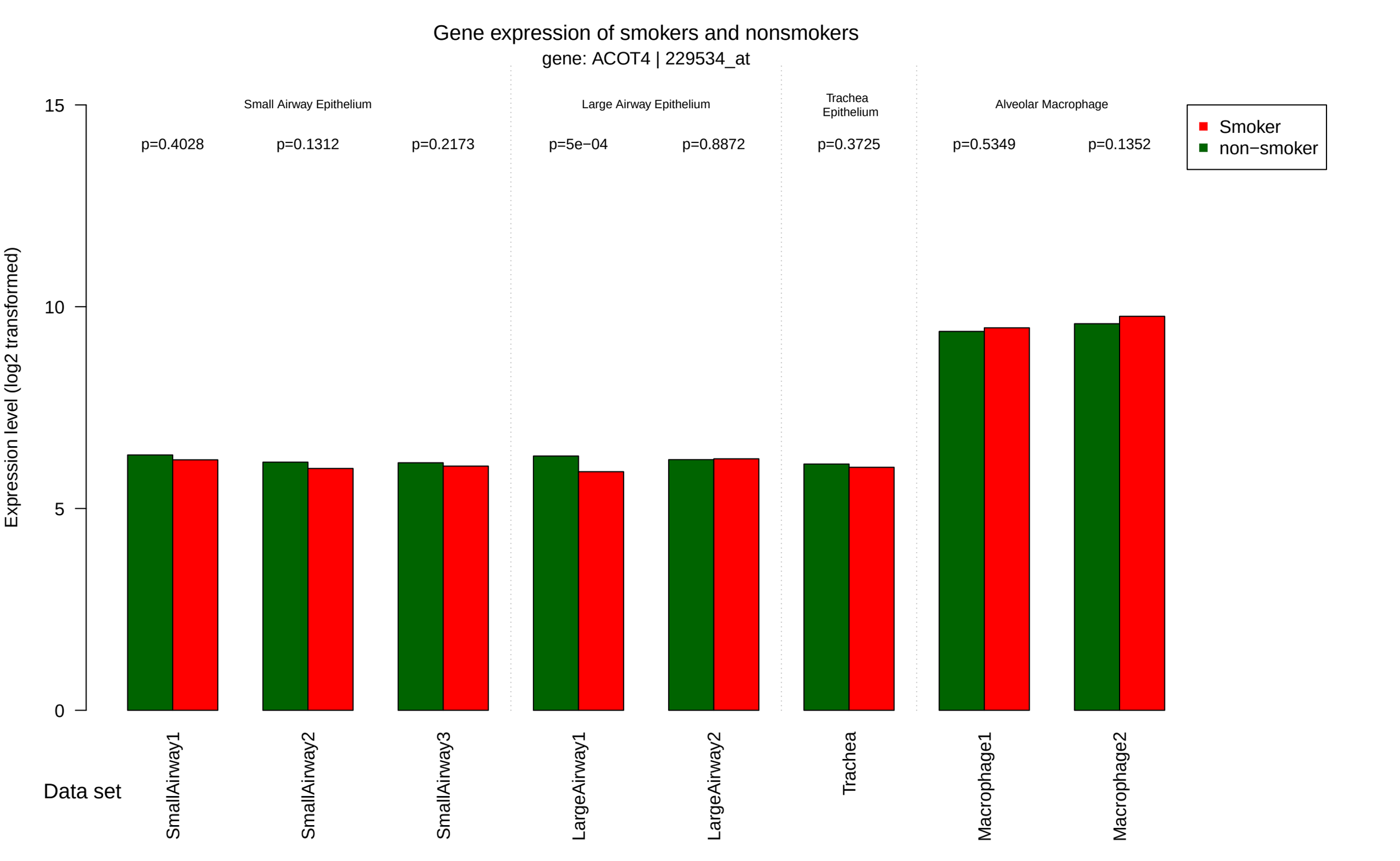
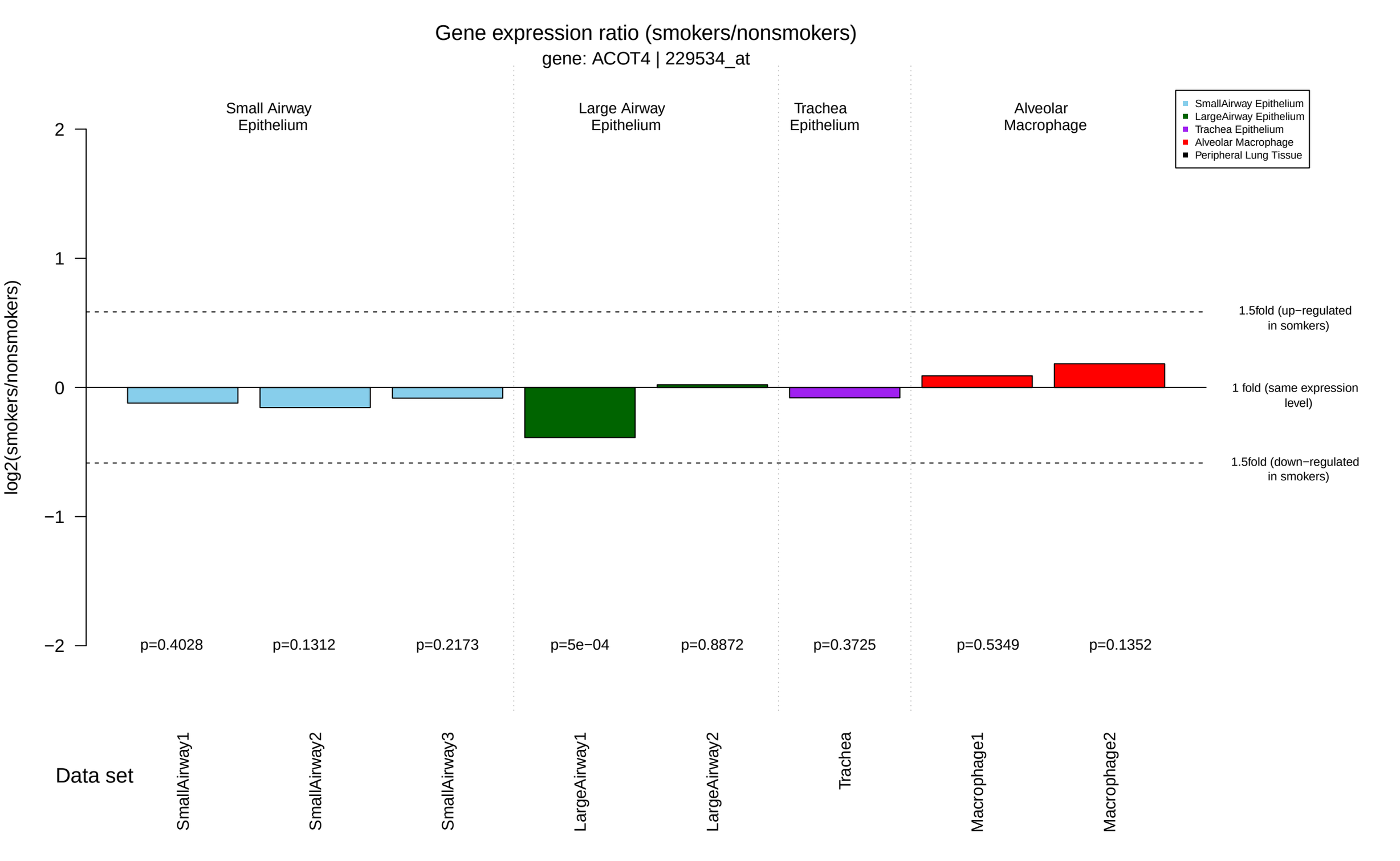
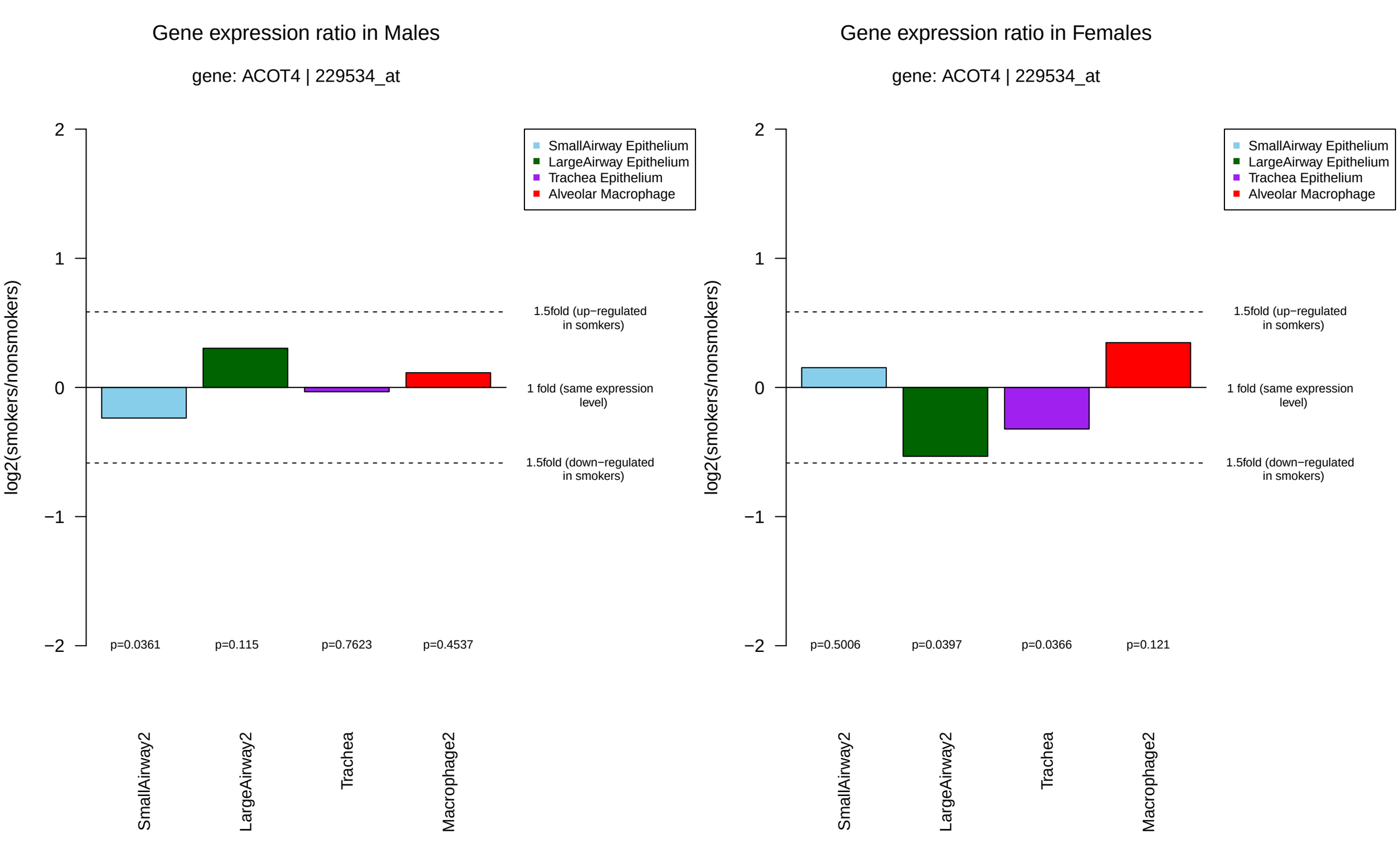
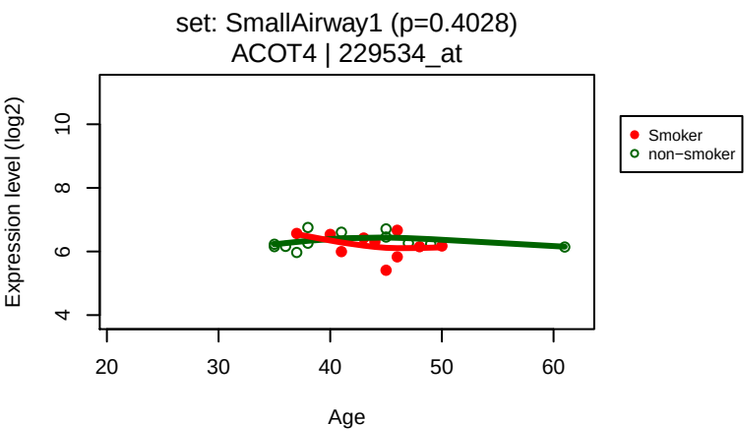
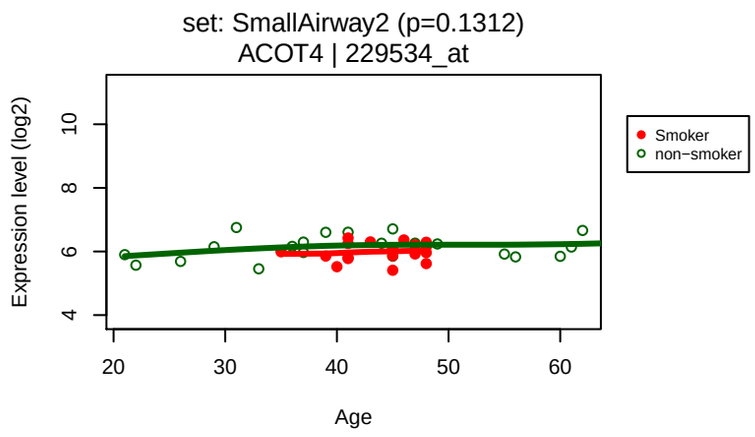
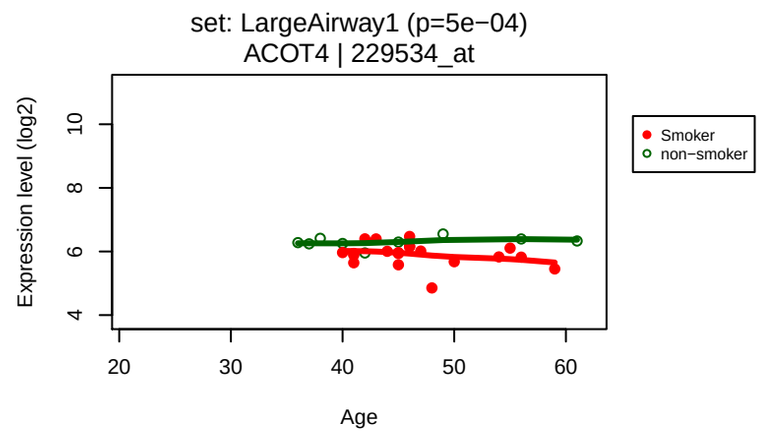

Gene expression of smokers and nonsmokers
gene: ACOT4 | 229534_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4028
p=0.1312
p=0.2173
p=5e−04
p=0.8872
p=0.3725
p=0.5349
p=0.1352
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

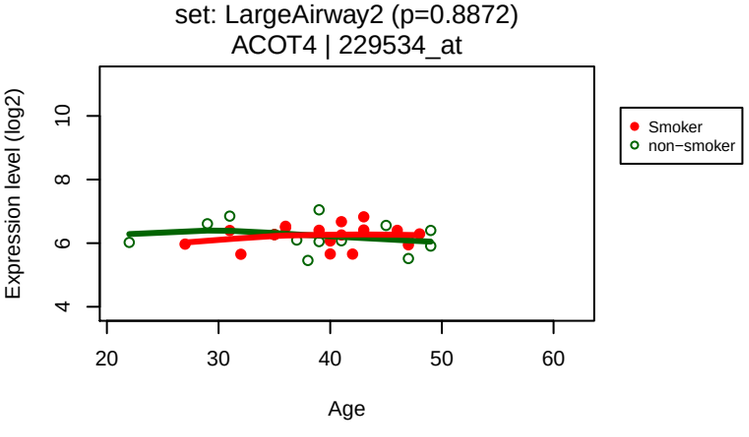

Gene expression ratio (smokers/nonsmokers)
gene: ACOT4 | 229534_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4028
p=0.1312
p=0.2173
p=5e−04
p=0.8872
p=0.3725
p=0.5349
p=0.1352
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

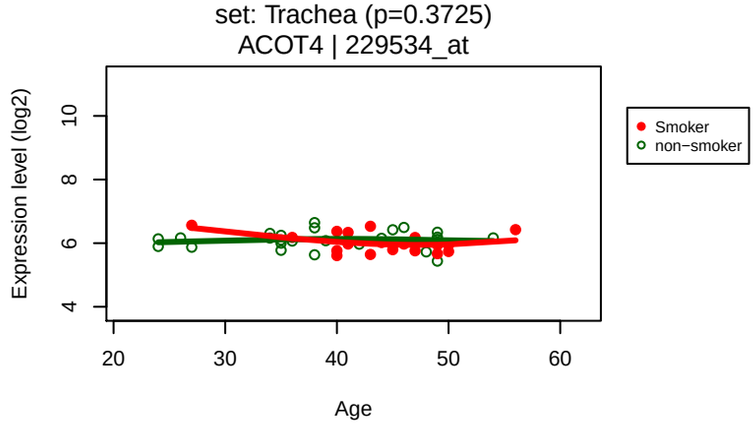

Gene expression ratio in Males
gene: ACOT4 | 229534_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0361
p=0.115
p=0.7623
p=0.4537
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACOT4 | 229534_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5006
p=0.0397
p=0.0366
p=0.121
SmallAirway2
LargeAirway2
Trachea
Macrophage2

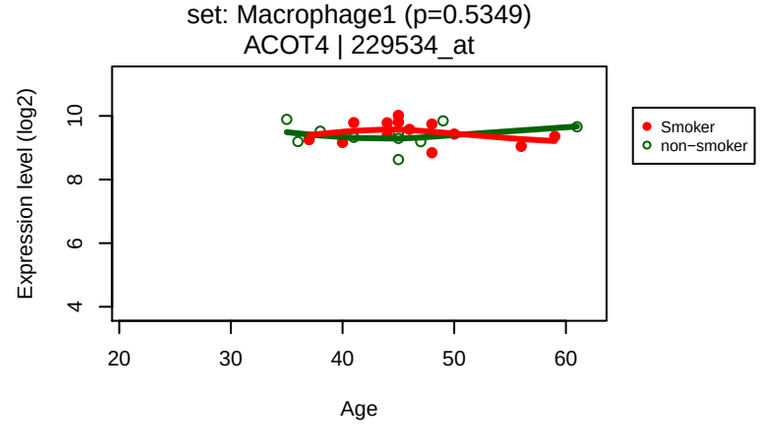

set: SmallAirway1 (p=0.4028)
ACOT4 | 229534_at
Expression level (log2)
Age
Smoker
non−smoker

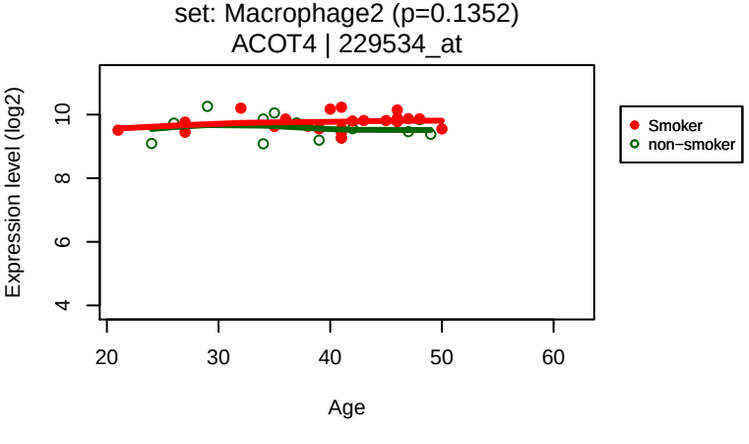

set: SmallAirway2 (p=0.1312)
ACOT4 | 229534_at
Expression level (log2)
Age
Smoker
non−smoker

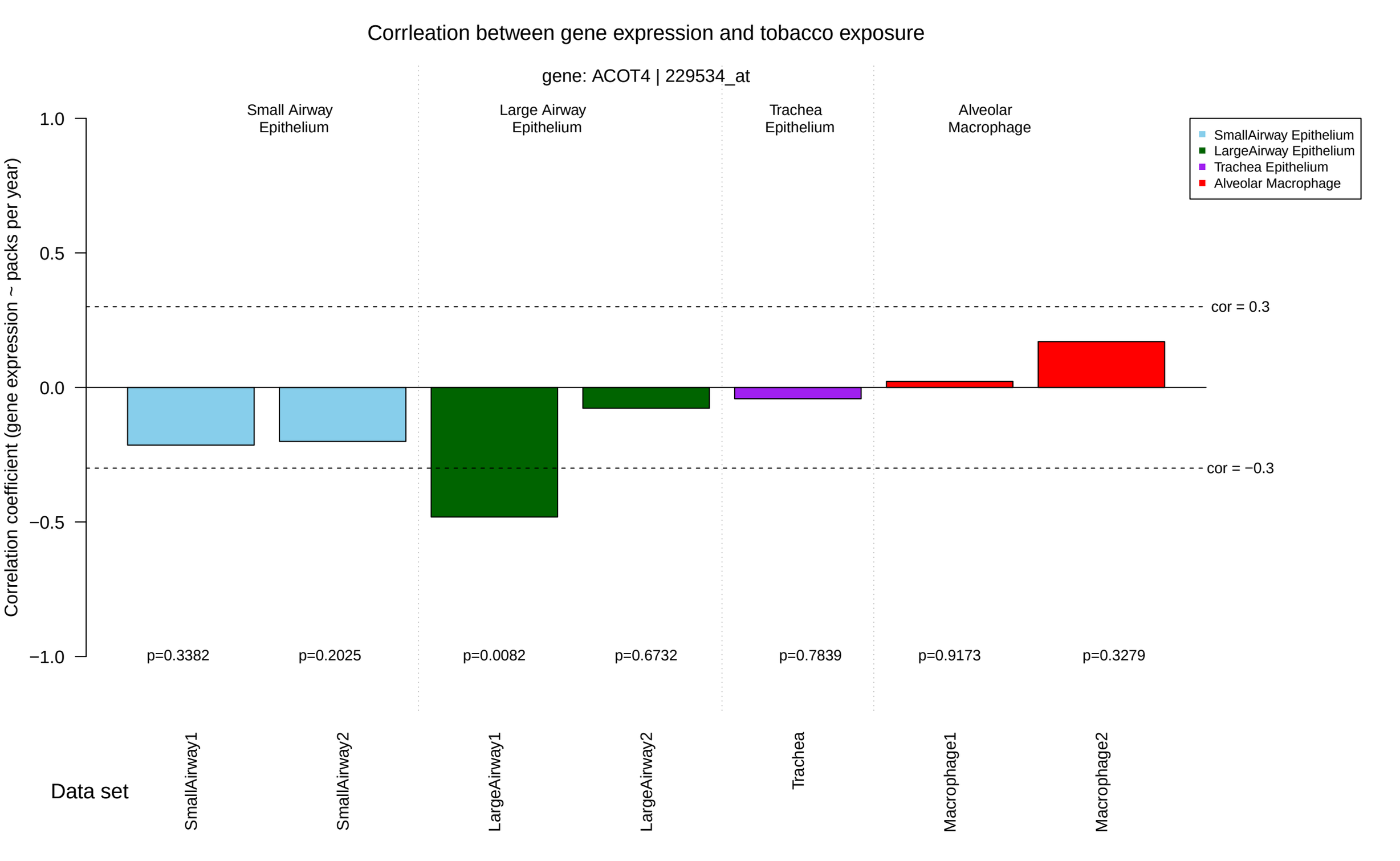

set: LargeAirway1 (p=5e−04)
ACOT4 | 229534_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8872)
ACOT4 | 229534_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3725)
ACOT4 | 229534_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5349)
ACOT4 | 229534_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1352)
ACOT4 | 229534_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACOT4 | 229534_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3382
p=0.2025
p=0.0082
p=0.6732
p=0.7839
p=0.9173
p=0.3279
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2